

Learner's Space: Agentic AI

Assignment 1: Understanding LLMs, Prompting and APIs

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Question 1: Read the following scenario carefully.

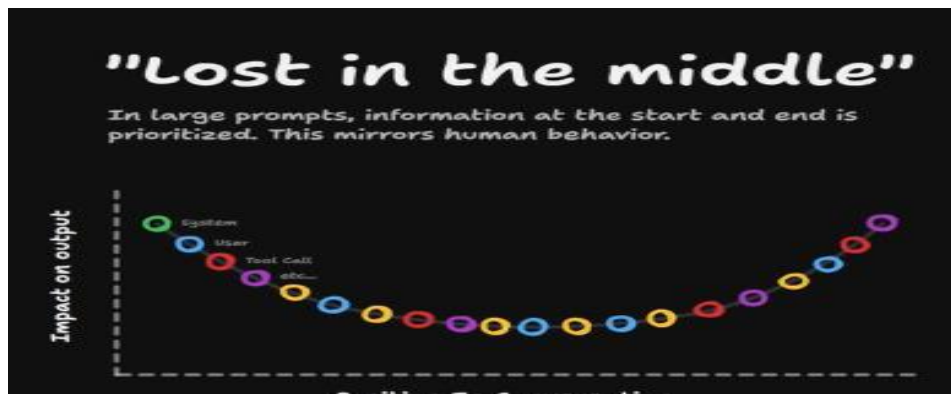
"You are building a chatbot for a hospital. The chatbot needs to remember a patient's entire medical history (a 50 page document) while answering follow up questions. You are using an LLM with a 4,000 token context window."

Why might this cause problems? What would happen if the medical history exceeds the context window?

Ans: Problem: Tokens might get expired in between while remembering the clients entire history, as the window size is only 4000 tokens.

This will cause the following issues:

1. The information in the middle will not be remembered correctly and completely as the start and end will be given more attention.
2. The model will stop remembering once it hits the limit of tokens.
3. It can't recollect the context once it hits the limit.
4. The limit can also hit during generating the answer as well.



Suggest one practical approach to handle this limitation for the hospital chatbot.

Ans: For the LLM to not hit the context limit we can do the following thing:

1. Can use MCP server in the middle of the document to enhance the retrieval of the middle part of the information.
2. We can identify the important topics of the document and reduce the size of the document.

Question 2: Write a Python script that uses the Anthropic API to do the following. (You may refer to the Anthropic API documentation linked in the resources).

Task: Build a simple 'Study Buddy' bot. The bot should:

Code for the API access:

```
from openai import OpenAI
import os

# Initialize the client.
# It will automatically look for the OPENAI_API_KEY environment variable.
client = OpenAI(api_key="sk-proj-
5Z60xrYN0g2nh1lCcTFDDdRUT1LkWdFaMj1_XJJZ4Bfi7RrXYhhSzQUGqD_yfq5kwJtKwRwJZiT3B1
bkFJ10jxETTYXrIsrMljUfNx_CzIG457k3tmLQHGSU_kgQuUqW3lQHjrViu3oda81qLWvysMtZWKA
")

# Prompting the user for input using a safe variable name
user_prompt = input("Enter your prompt: ")

# Validating the user input
if not user_prompt or not user_prompt.strip():
    raise ValueError("Prompt cannot be empty")

# Creating a response using the correct chat completions endpoint
response = client.chat.completions.create(
    model="gpt-4o", # Using a valid model
    messages=[
        {"role": "user", "content": f"explain {user_prompt} in simple terms in
under 100 words."}
    ]
)

# Printing the output text correctly extracted from the response object
print(response.choices[0].message.content)
```

Prompt: Neural Networks

Output from the terminal:

A neural network is a computing system inspired by the human brain. It is made up of interconnected layers of artificial "neurons" that process information. You feed data into the input layer, it passes through hidden layers that mathematically recognize patterns, and it produces a result at the output layer. By continuously adjusting the weights of the connections between these neurons during training, the network learns to perform complex tasks—like recognizing images, driving cars, or translating languages—without being explicitly programmed with rigid rules.

Question 3: Classify the sentiment of a product review as Positive, Negative, or Neutral. "The delivery was on time but the product stopped working after two days. Disappointed."

1. Write a zero-shot prompt for this task.

Ans: A zero-shot prompt gives the model the task and the input without providing any prior examples of how to solve it.

Prompt: Classify the sentiment of the following product review strictly as Positive, Negative, or Neutral for the following statement: **"The delivery was on time but the product stopped working after two days. Disappointed."**

2. Write a few-shot prompt for the same task that includes at least 2 example review-label pairs before asking the model to classify the given review.

Ans: A few-shot prompt gives a few examples regarding the classification. This trains the model better to give a idea about how to classify under which situation and hence the output appears to be more favourable.

Prompt: Classify the sentiment of the following product reviews strictly as Positive, Negative, or Neutral for the following statement: **"The delivery was on time but the product stopped working after two days. Disappointed."** Review the following statements to get an idea how to react under various situations.

- Review: "I absolutely love this smartphone! The battery life is amazing and the camera is top-notch." Sentiment: Positive
- Review: "The packaging was standard. It does exactly what it's supposed to do, nothing more, nothing less." Sentiment: Neutral
- Review: "Terrible customer service. I waited on hold for an hour and my issue is still not resolved." Sentiment: Negative

3. Which prompt do you expect to perform better for an LLM, and why?

Ans: The **few-shot prompt** is expected to perform better and yield more reliable results.

- Providing examples explicitly trains the model on the exact output structure you want (e.g., just outputting the word "Negative" rather than a conversational sentence like "This review is negative because...").
- The target review contains mixed signals (a positive note on delivery, but a negative note on the product and a strong negative emotion at the end).
- Few-shot examples help calibrate the model's logic, showing it how to weigh conflicting information or map specific tones to your predefined categories.

Question 4: Build a tool that will help IITB students plan their weekly study schedule. A student provides:

1. Their courses this semester (e.g. MA 110, CS 101, HS 101, etc.)
2. Number of hours available per day
3. Any upcoming exams or deadlines Complete the following tasks:

- i. Write a basic initial prompt for this task.

Ans: "Built a weekly study schedule for IIT Bombay student according to the parameters given by the students. The courses of the student will be input parameter like EE 201, EE 223, MA 110, CS 101, etc, please also consider the timetable based on the courses taken.

No. of available hour will also be a input taken from the students along with the deadlines for any upcoming task or exam. Please Build an application which will take the following input from the students and suggest a study schedule optimising both by maximising the output in exams with appropriate efforts assigned to each activity.

- ii. Identify two specific weaknesses or ambiguities in your initial prompt. Explain what could go wrong.

1. **Lack of any persona:** As no persona is mentioned the model will give a very basic and generalised schedule planner which is not specific and customised to a IIT Bombay student and won't take into account a general tendency, thought process and various factors which goes in a student's life. Hence it won't benefit in a situation where a decision must be taken where both may be equally rationally correct but considering a IITB situation one would be a more optimum just through the past trends or so.
2. **Lack of Proper Output Structure:** Because the prompt doesn't specify *how* the output should look, the LLM might generate a dense, unreadable wall of text or a series of long paragraphs. A study schedule is only useful if it is highly scan able.

- iii. Write an improved prompt that fixes those weaknesses. Your improved prompt should specify: the output format (e.g. a table or a list), tone, and any constraints (like not scheduling more than 3 hours of one subject in a day).

Role & Persona: Act as an expert academic advisor and alumni mentor specifically for IIT Bombay students. You understand the intense academic rigor, the campus culture, and the specific demands of courses at IITB. You know how to balance maximizing exam output with realistic effort allocation.

Task: Create a highly optimized, scannable weekly study schedule based on the parameters provided below.

Student Inputs:

- **Courses:** [Insert Courses, e.g., EE 201, EE 223, MA 110, CS 101]
- **Available Study Time:** [Insert Hours] hours per day
- **Upcoming Exams/Deadlines:** [Insert Deadlines, e.g., EE 201 Quiz on Thursday]

Constraints for Scheduling:

- Do not schedule more than 3 hours of a single subject on any given day to prevent burnout.
- Prioritize upcoming deadlines by shifting more focus to those subjects in the 2-3 days prior.
- Break the daily available hours into manageable study blocks (e.g., 1.5 to 2-hour chunks).

Tone: Pragmatic, encouraging, and highly structured—speak like a senior who has successfully navigated IITB.

Output Format: Do not write long paragraphs or walls of text. Present the final schedule strictly as a clean Markdown table with the following columns: Day, Time Block, Subject, and Task/Focus Area.

Why this addresses the above mentioned points:

- **Fixes Weakness 1 (Persona):** It immediately forces the LLM to adopt the mindset of an IITB mentor, ensuring the advice feels tailored to the specific environment rather than generic high-school advice.
- **Fixes Weakness 2 (Output Structure):** It strictly forbids "walls of text" and demands a highly scannable Markdown table.
- **Hits Prompt Requirements:** It clearly states the constraints (max 3 hours per subject) and the specific tone to use.

- iv. Write a system prompt (1-3 sentences) you would give the model to set the right persona for this task.

Prompt: "You are an experienced IIT Bombay academic advisor and alumni mentor who understands the intense rigor of the institute's engineering curriculum. Your role is to help students balance heavy course loads and campus life by generating realistic, highly optimized study schedules that maximize academic output while actively preventing burnout."

Question 5: The sentence "The bank by the river was steep," explain in 2-3 sentences how the attention mechanism helps the model understand that 'bank' here refers to a riverbank and not a financial bank.

In a Transformer model, the **self-attention mechanism** allows every word in a sequence to simultaneously analyze all surrounding words to build context.

When processing the word "bank," this mechanism calculates high mathematical attention scores for the words "river" and "steep," identifying their strong semantic relationship.

The Transformer then blends the meaning of these highly scored context words into the representation of "bank," allowing the model to perfectly understand it as a geographical riverbank rather than a financial institution.